Task E2 — Ansible Deployment Playbook

logibridge_deploy.yml: 7 tasks, run twice | LogiEdge | FreightBridge Logistics Pvt. Ltd. | Component E, Module 5

At a glance

Item	Result
Playbook	`logibridge_deploy.yml`: one play, exactly 7 tasks in the required order
Second run	Designed to report **changed=0**: the stop task runs only when a file changed, the pull uses a version tag, and the start task is declarative
Verified here	Structure, Jinja templates, and the two shell tasks run against a fake docker command (21 tests pass)
Not verified	**The playbook has not been run by Ansible**, and no changed=0 output exists yet: Ansible and Docker were not available to me. Run `./run_playbook_twice.sh` and record it (Section 5).

1 The seven tasks

#	Task	Module	Reports “changed” when	Second run
1	Create /opt/logibridge	file	the directory is missing	ok
2	Copy model to /opt/logibridge/model.tflite	copy	the file content differs (checksum)	ok
3	Copy reference_dist.json	copy	the file content differs (checksum)	ok
4	Stop the container if running	shell	a running container was stopped; runs only if task 2 or 3 changed	skipped
5	Pull the image from the local registry	docker_image	the tag was not on the truck yet	ok
6	Start the container with environment variables	docker_container	it was stopped, missing, or its image or settings differ	ok
7	Wait 15 seconds and verify it is running	shell	never (fails if the container is not running)	ok

The container receives MODEL_PATH, REFERENCE_DIST_PATH, DRIFT_SCORE, TRUCK_ID, MQTT_HOST and MQTT_PORT, and mounts /opt/logibridge read-only, so a new model file takes effect with no image rebuild (the Task D2 MODEL_PATH feature) and the Task E1 monitor reads the same reference file.

2 Why a straightforward playbook fails the second-run test

The obvious version stops the container (task 4), pulls (task 5) and starts it (task 6) on every run. A stop task that always stops, followed by a start that always starts, reports two changes on a run where nothing changed. Three decisions avoid that:

Decision	Reason
Task 4 is a shell command that stops only a container that is running, and runs only when the model or reference file changed	Ansible's docker_container with state=stopped would create a missing container first (documented behaviour), which is wrong on a first deployment. The shell version reports changed only if it stopped something.
Task 5 pulls a version tag (v1, v2) instead of latest	A tag already on the truck is not pulled again, so repeat runs change nothing. A new image is rolled out by changing image_tag. Set force_pull=true only for a mutable tag.
Task 6 declares the end state (started, with this image and environment)	Ansible then does nothing if it already matches, starts it if task 4 stopped it, and recreates it if the image differs, so no separate restart is needed.

3 What to expect

	First deployment	Second run (no changes)	New model file	New reference file only	New image tag
1 Create /opt/logibridge	**changed**	ok	ok	ok	ok
2 Copy model.tflite	**changed**	ok	**changed**	ok	ok
3 Copy reference_dist.json	**changed**	ok	ok	**changed**	ok
4 Stop container if running	ok	skipped	**changed**	**changed**	skipped
5 Pull image from registry	**changed**	ok	ok	ok	**changed**
6 Start container with env	**changed**	ok	**changed**	**changed**	**changed**
7 Wait 15 s, verify running	ok	ok	ok	ok	ok

Predicted from the playbook's conditions, not observed. Second-run column outlined: changed=0 on every task.

Run	Expected PLAY RECAP line (not observed)
First	ok=7 changed=5 unreachable=0 failed=0 skipped=0
Second, no changes	**ok=6 changed=0 unreachable=0 failed=0 skipped=1**

The first run is expected to change tasks 1, 2, 3, 5 and 6 on a clean machine. The second run takes at least 15 seconds because of task 7. If your recap differs, send it to me.

4 Running it

```
# once: tools, local registry, model and reference
pip install ansible docker && ansible-galaxy collection install community.docker
python3 convert_to_tflite.py                        # model.tflite (Task D2)
docker build -t logiedge/inference:v1 . && ./prepare_demo_registry.sh   # localhost:5000/logiedge/inference:v1

# the demo: both runs, then the changed=0 check
./run_playbook_twice.sh              # add -K if sudo asks for a password, or -e use_become=false

# a real update later
cp new_model.tflite model.tflite && ansible-playbook -i inventory.ini logibridge_deploy.yml
```

5 Demo video script

Time	On screen	Say
0:00	logibridge_deploy.yml, scrolling the 7 task names	The playbook has exactly seven tasks, from creating the directory to verifying the container.
0:30	`./run_playbook_twice.sh`, run 1 recap	The first run deploys: the directory, model, reference and image are new, so five tasks change.
1:10	Run 2 starts; the 15-second wait	Now I run it again with nothing changed in between.
1:40	Run 2 recap and the PASS line	The second run reports changed=0. The stop task is skipped because no file changed, and nothing else needs to change.
2:00	docker ps; docker logs showing PSI lines	The container is running with the new model and the drift monitor from Task E1.

6 Verification

Check	How	Result
One play, exactly 7 tasks, required names and order; no handlers, roles or includes	Tests	Pass
Correct modules; paths /opt/logibridge, model.tflite, reference_dist.json; pull from localhost:5000; environment variables	Tests	Pass
The container's configuration reader accepts the environment the playbook sets	Tests	Pass
Jinja templates render, including Docker's {{.State.Running}} inside a raw block	Rendered with jinja2	Pass
Stop guard: skipped when no file changed; changed only when a container was actually stopped	Condition expressions evaluated	Pass
Stop and verify shell tasks: stop a running container, do nothing when stopped or missing; verify fails unless running	Real scripts, fake docker	Pass
Module behaviour (docker_container stopped, image-change recreate, docker_image pull) matches the documentation	Read in the collection docs	Checked
Ansible run, real Docker, changed=0 on the second run, the video	Needs Ansible and Docker	Not run

7 Items to confirm

- **Image tag.** Pulling a version tag is what keeps the second run at changed=0. If you want “latest”, use `-e force_pull=true`; I have not tested whether your collection version then reports changed=false for an unchanged image.
- **Task 4 runs only when a file changed.** The brief says “stop the running container if running”. Always stopping would make the second run report changes, so I guarded it. Tell me if you want it unguarded and accept changed=2.
- **Privileges.** The playbook uses sudo (`become`). On a developer machine add `-K` or `-e use_become=false`.
- **Networking.** The container uses host networking so MQTT_HOST=localhost reaches the truck's Mosquitto; change mqtt_host if your broker is elsewhere.
- **Prerequisites on the target:** Docker, the Python docker package and the community.docker collection.